

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 11 PRELIMINARY COURSE

Extension 1 Mathematics

Task 1

May 2011

TIME ALLOWED: 70 minutes

Instructions:

- Write your name and class at the top of this page, and on all your answer sheets.
- Hand in your answers attached to the rear of this question sheet.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied at the time of marking
- *Start each new question on a new page*

(FOR MARKERS USE ONLY)

1	2	3	4	5	6	TOTAL
/9	/9	/9	/9	/9	/9	/54

QUESTION 1 (9 Marks):**Marks**

- (a) Fully factorise and simplify:

2

$$\frac{6^n - 2^n}{9^n - 3^n}$$

- (b) Show that the expression $\frac{1}{3\sqrt{2}-2} - \frac{1}{3\sqrt{2}+2}$ is rational.

2

- (c) What is the Domain of the function given by $y = \frac{1}{\sqrt{x^2-1}}$?

2

- (d) If $\sin \alpha = x$ and $90^\circ \leq \alpha \leq 270^\circ$, find an expression for $\tan \alpha$

3**QUESTION 2 (9 Marks):** *(Start a New Page)*

- (a) On a set of Cartesian axes, neatly graph $|x| + |y| = 1$, showing all intercepts on the axes

2

- (b) (i) Sketch the graph of $y = \sqrt{9 - x^2}$

2

- (ii) Give the Domain and Range of the function

2

- (iii) On your diagram, neatly shade the solution to $y < \sqrt{9 - x^2}$, carefully indicating whether boundaries are included or excluded from the solution.

3

QUESTION 3 (9 marks): (Start a New Page)

- (a) Solve the two equations below simultaneously to find their point(s) of intersection:

3

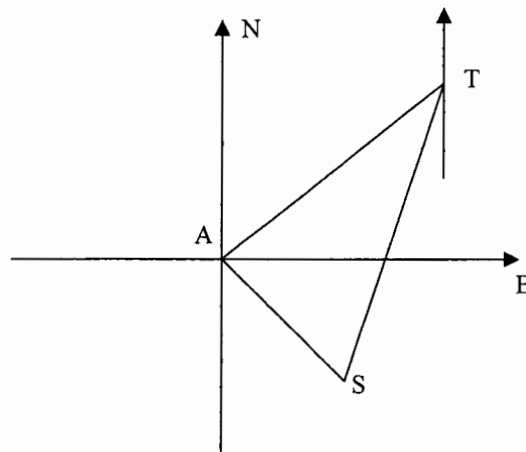
$$xy = 6$$

$$y = x - 5$$

- (b) The diagram below represents the information given in the following problem:

A ship S sails from a port A on a bearing of $S18^\circ E$ at a speed of 10 km per hour.

A second ship, T, leaves at exactly the same time on a course of $N72^\circ E$. After 2 hours the ship T sights S on a bearing of $S24^\circ W$



- (i) Copy the diagram onto your answer page and put on it all the information given in the above question
- (ii) Find the size of angle ATS
- (ii) Find the distance between the two ships 2 hours after leaving port A (ie ST). (to the nearest km)

2

1

3

You must show all necessary working to gain full marks

QUESTION 4 (9 Marks): *(Start a New Page)*

Marks

- (a) (i) *On the same set of axes*, sketch the graphs of $y = |2x - 1|$ and $y = 1 - 2x$

3

- (ii) Use the diagram, or otherwise, to solve

$$|2x - 1| > 1 - 2x$$

for all real x .

- (b) Solve, for $0^\circ \leq \theta \leq 360^\circ$,

3

$$4\cos^2\theta - 3 = 0$$

- (c) Prove that $\frac{\cos x}{1 + \sin x} + \frac{1 + \sin x}{\cos x} = \frac{2}{\cos x}$

3

QUESTION 5 (9 Marks): *(Start a New Page)*

- (a) You are given the function $f(x) = \frac{2}{x^2 - 4}$

- (i) Is the function odd, even, or neither? (Give reasons)

1

- (ii) What are the vertical asymptotes for $f(x)$?

1

- (iii) Where does this curve cut the y -axis?

1

- (iv) Sketch the curve $y = f(x)$, showing all important features and indicating the **horizontal** asymptote(s).

4

- (b) Find all solutions over the Real number system to:

2

$$\frac{1}{x-1} < \frac{1}{x}$$

Show all necessary working

QUESTION 6 (9 Marks): (Start a New Page)

(a) Simplify $\frac{\cos(90^\circ - \theta)}{\sin(90^\circ + \theta)}$ 2

(c) A function is defined by the following:

$$f(x) = \begin{cases} 3, & \text{if } x \leq -2 \\ -x, & \text{if } -2 < x < 0 \\ x^2, & \text{if } x \geq 0 \end{cases}$$

(i) Sketch $y = f(x)$ for $-4 \leq x \leq 2$ 3

(ii) Find the value of $f\left(\frac{1}{2}\right) + f\left(-\frac{1}{2}\right)$ 1

(iii) Find the value of $f(|a|)$ where a is real. 1

(c) $g(x) = 2x + 4$ is a function for all real values of x . 2

A second function, $f[x]$ is defined thus:

$$f[g(x)] = x$$

Find the function $f[x]$

END OF EXAMINATION PAPER

SOLUTIONS - Maths Extension 1

YEAR 11 - EXT

May 2011

① (a) $\frac{2^n(3^n-1)}{3^n(3^n-1)} = \left\{ \begin{array}{l} \frac{2^n}{3^n} \\ (\frac{2}{3})^n \end{array} \right.$

1 mark for each factorisation
= ②

(b) $\frac{3\sqrt{2}+2}{14} - \frac{3\sqrt{2}-2}{14}$

1 mark for the rationalisation

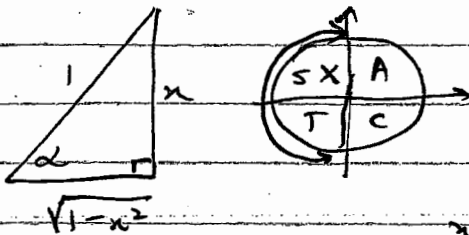
= $\left\{ \begin{array}{l} \frac{4}{14} \\ \frac{2}{7} \end{array} \right.$ which is rational

1 mark for $\frac{4}{14}$ or $\frac{2}{7}$

(c) Q: $x \geq 1$ or $x < -1$

② MARKS - subtract 1 if they have "=" signs.

(d)



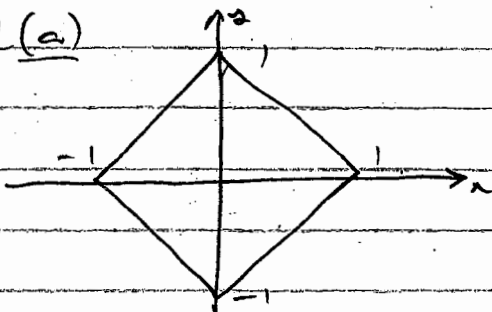
$\tan \alpha = -\frac{x}{\sqrt{1-x^2}}$

1 mark for $\sqrt{1-x^2}$

1 mark for the expression for $\tan \alpha$

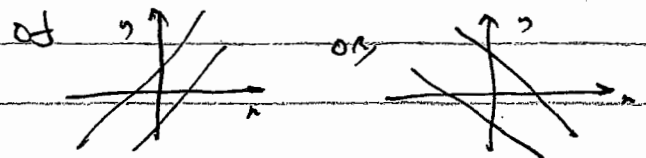
1 for the negative

2(a)

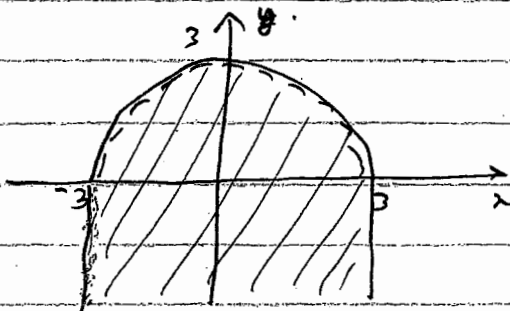


2 for the graph and its intercepts

1 MARK ONLY for either



(b)



② MARKS - 1 for the shape and 1 for intercepts. (1 only if full circle)

(ii) Q: $-3 \leq x \leq 3$ R: $0 \leq y \leq 3$

1 mark each

(iii)

1 mark for shading inside the circle
1 mark for the extension
1 mark for dots and heavy line

(3)(a)
$$\begin{cases} y = \frac{6}{x} \\ y = x - 5 \end{cases}$$

(1) into (2) $\frac{6}{x} = x - 5$

$$6 = x^2 - 5x$$

$$x^2 - 5x - 6 = 0$$

$$(x - 6)(x + 1) = 0$$

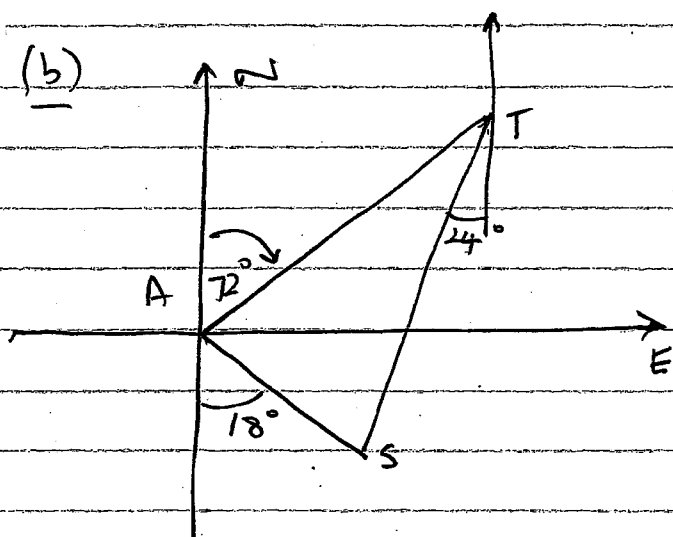
$\therefore x = 6$ or $x = -1$

$\therefore$ Points are $(6, 1)$ and $(-1, -6)$

← 2 MARKS

← 1 MARK

(b)



All of these angles must be shown to get 2 MARKS

(no extra marks for $\angle TAS = 90^\circ$ or $AS = 20$)

(c)
$$\angle ATS = (72 - 24)^\circ = 48^\circ$$

← 1 MARK

(ii) $AS = 20$ and $\angle TAS = 90^\circ$

① MARK for both

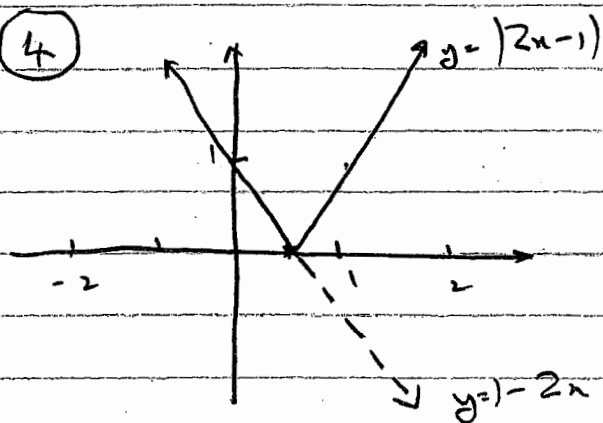
$$\therefore \frac{ST}{20} = \frac{1}{\cos 48}$$

← 1 MARK

$$\therefore \left. \begin{array}{l} ST = 29.88 \\ \approx 30 \text{ km} \end{array} \right\}$$

1 for either

(4)



← 1 MARK.

= (3)

← 1 MARK.

$$x > \frac{1}{2}$$

← 1 MARK

(b)

$$\cos \theta = \pm \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$$

← (1) MARK

← (2) MARKS. 1 only
for 30° , or for $30^\circ, 150^\circ$

(c)

$$\text{LHS} = \frac{\cos x (1 - \sin x)}{1 - \sin^2 x} + \frac{(1 + \sin x) \cos x}{\cos^2 x}$$

$$= \frac{\cos x (1 - \sin x) + (1 + \sin x) \cos x}{\cos^2 x}$$

$$= \frac{2}{\cos x}$$

There are several methods.

But for 3 marks, they

cannot mark down both

sides (2 max.) and

must show enough steps

to explain what is happening

5 (a) $f(x) = \frac{2}{x^2-4}$

(i) $f(a) = \frac{2}{a^2-4}$ $f(-a) = \frac{2}{(-a)^2-4}$

$= \frac{2}{a^2-4}$
 $= f(a)$

$\therefore$ EVEN.

1 MARK but
 some explanation as to
why should be given
 (may not be done as
 this)

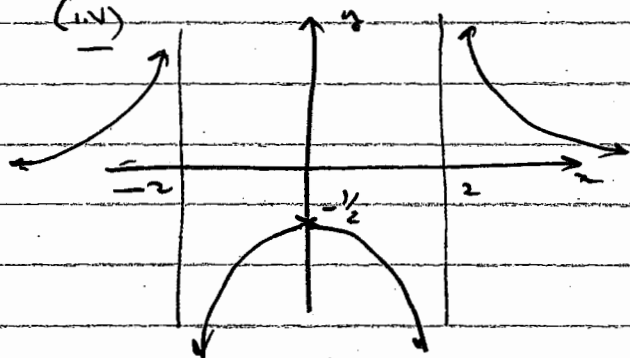
(ii) $x \neq \pm 2$

← 1 MARK

(iii) Cuts y-axis at $(0, -\frac{1}{2})$

← 1 MARK

(iv)



1 MARKS for each arm = (2)

① for the part between
 $-2 < x < 2$

① for indicating $y=0$ as
 the asymptote.

(b) $\frac{1}{x-1} < \frac{1}{x}$

C.V. $x \neq 0$
 $x \neq 1$

$x-1 = x$

NO SOLN



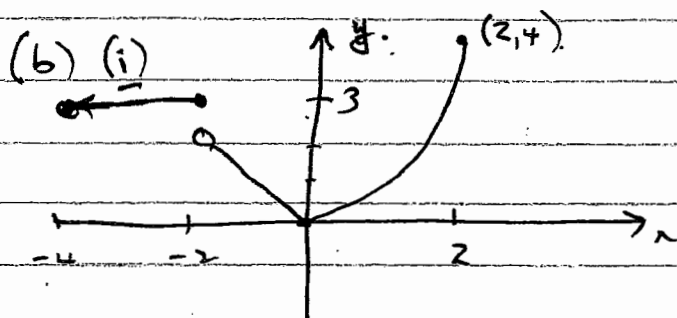
$0 < x < 1$

1 for any accurate
 method shown.

← 1 for solution

(6) (a) $\frac{\cos(90^\circ - \theta)}{\sin(90^\circ + \theta)} = \frac{\sin \theta}{\cos \theta}$
 $= \tan \theta$

(2) 1 for each simplification



1 MARK for each "part"

= (3)

Subtract 1 MARK for no open circle

Subtract 1 if the graph doesn't "end" at both ends

(ii) $f\left(\frac{1}{2}\right) + f\left(-\frac{1}{2}\right) = \frac{1}{4} + \frac{1}{2}$
 $= \frac{3}{4}$

1 MARK

(iii) Since $|a| \geq 0 \quad \forall a$,
 $f(|a|) = a^2$

1 MARK

(c) $f[x] = \frac{x-4}{2}$

(2) MARKS

1 for division by 2

1 for subtraction of 4